

Alexander Ticket

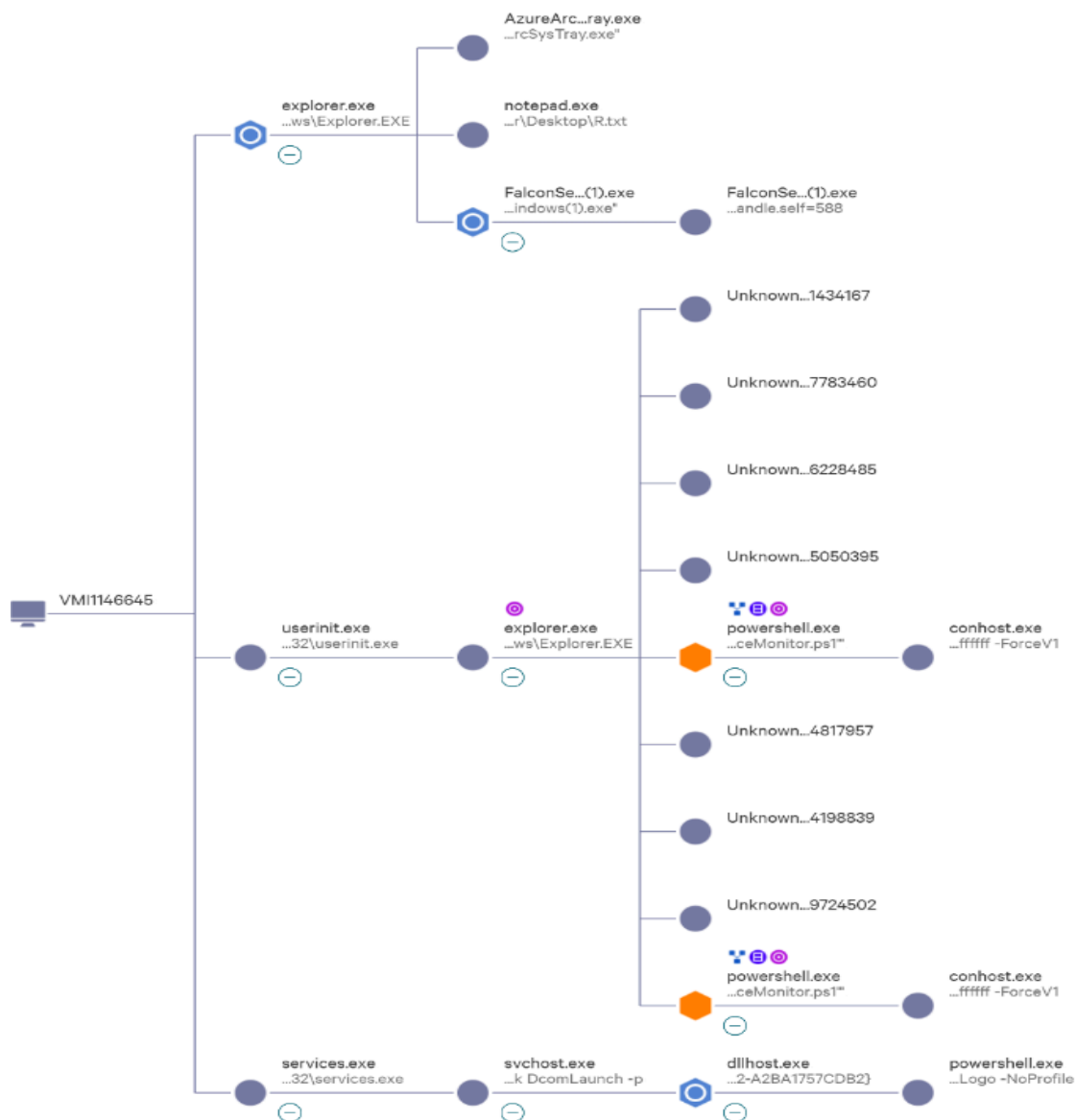
Note: This report has been sanitized for public sharing.

All internal IPs, hostnames, and CrowdStrike URLs have been redacted or replaced with simulated values.

Report was originally prepared for Jira; internal console links are not publicly accessible. Query references shown for context

The Summary :

The alert indicates a sequence of debugger evasion, Falcon sensor impairment, and malicious PE execution via RDP on host [redacted-host], confirmed by CrowdStrike Intelligence indicators.



CrowdStrike Link:

- **Incident(CrowdScore):**

(internal link, redacted for confidentiality)

- **Endpoint Detection:**

(internal link, redacted for confidentiality)

Description

A multi-stage intrusion was detected on host [redacted-host], characterized by the execution of a suspicious PowerShell script and targeted efforts to compromise system defenses. The activity originated from the execution of **PerformanceMonitor.ps1** located in the Administrator's Downloads directory. The script utilized an execution policy bypass (-Scope **Process Bypass**) to facilitate the running of unsigned code, a technique frequently observed in defense evasion.

During the active RDP session, a malicious PE file was written and executed, suggesting a manual, human-operated attempt at lateral movement. Advanced telemetry identified a specific attempt to impair system security where a process opened a handle to a Falcon process with excessive permissions requested. This effort was coupled with defense evasion via debugger checks, intended to detect and avoid tracing or analysis by security tools.

Network analysis confirmed a domain lookup for **shichizaoa.top**, which directly matches CrowdStrike Intelligence indicators for domains previously leveraged in targeted attacks. The sequence of RDP-based payload delivery, the attempt to neutralize endpoint protection, and the communication with known malicious infrastructure indicates a coordinated intrusion aimed at establishing a foothold and maintaining persistence.

◆ Host Information:

Hostname: [redacted-host]
Operating System: Windows Server 2022
IP Address:[redacted internal IP]
Local IP Address:[redacted]
Host Type: Server

◆ User Information:

1. Initial Compromised Account:

Username: **Administrator**
Local Admin: **Yes**
Role: **Used for initial RDP access**

Login Time: Feb. 2, 2025 09:39:41

Login Domain: [internal-domain-redacted]

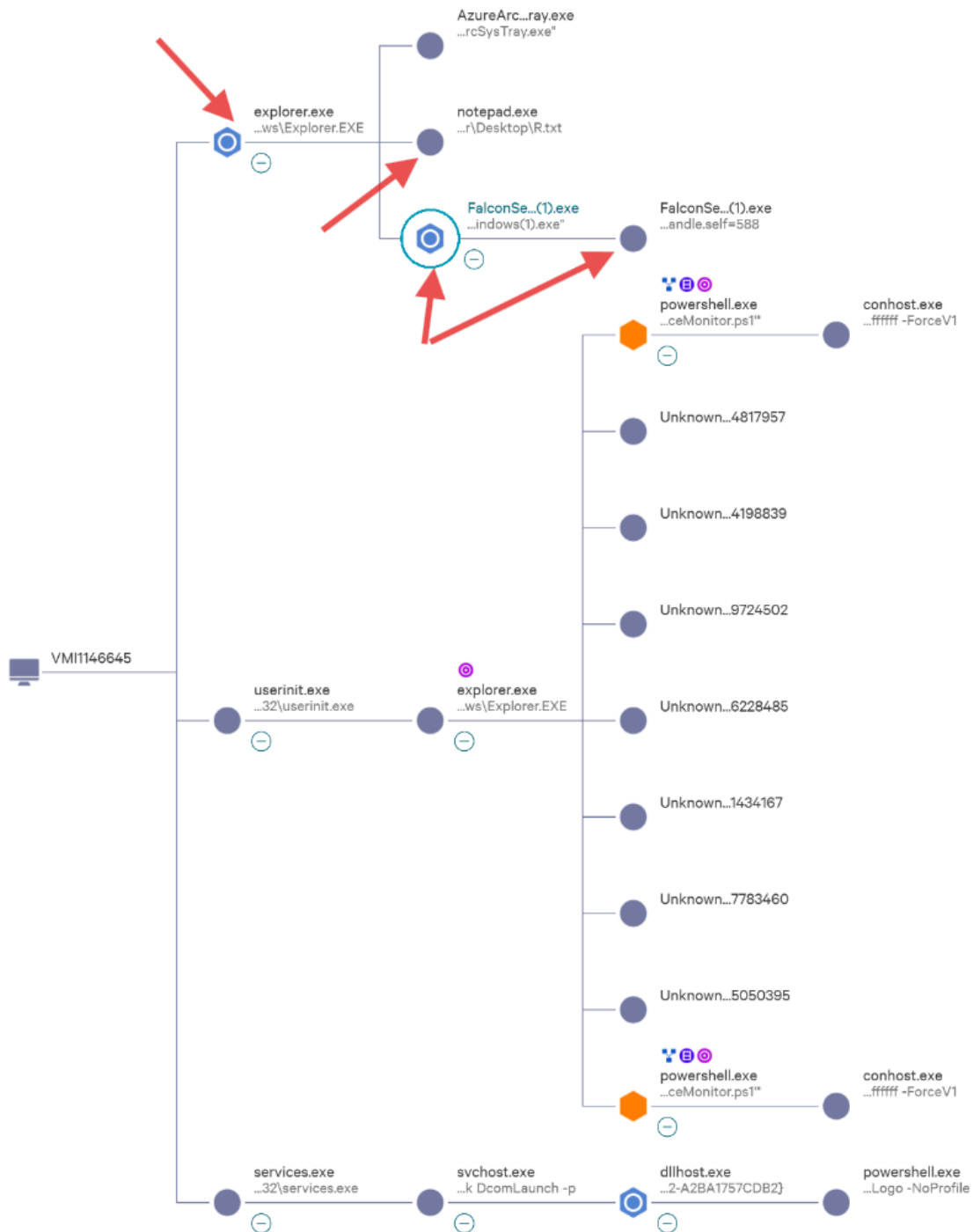
◆ Tactic & Technique:

Execution via Command and Scripting Interpreter, Lateral Movement via Remote Desktop Protocol, Defense Evasion via Debugger Evasion, Defense Evasion via Impair Defenses, Malware via Malicious File, Falcon Intel via Intelligence Indicator - Domain

MITRE ATT&CK ID: T1059.001, T1021.001, T1622, T1562.001

Investigation Findings and Analysis:

Observed Command 1:



① Phase 1 : Interactive Session and Defense Impairment Analysis

Process: explorer.exe

Start time: Dec. 16, 2024 11:30:10

Executable SHA256

53f36699c35c8f2360608a79f0809ba888c61f15886ae2b1f209a3e9b896cba7

While the process started on Dec. 16, 2024, it triggered a critical automated lead on Feb. 2, 2025, at 06:51:55. The process requested a handle to the Falcon sensor process with **excessive permissions**. This behavior is classified as **Defense Evasion via Impair Defenses**. It indicates a deliberate attempt to gain unauthorized access to the security agent's internal memory or configuration to potentially disable monitoring



Identified by CrowdStrike as an automated lead:

Detection: Defense Evasion via Impair Defenses

Detection Timestamp: Feb. 2, 2025 06:51:55

Command line

C:\Windows\Explorer.EXE

File path

\Device\HarddiskVolume2\Windows\explorer.exe

Identified by CrowdStrike as an **Automated Lead** for **Defense Evasion via Impair Defenses**.

Although no direct block was triggered, the process was flagged for requesting excessive permissions to the Falcon sensor, indicating a targeted attempt to blind the EDR.

Process: notepad.exe

Start time: Feb. 2, 2025 06:50:37

End time: Feb. 19, 2025 07:47:20

Command line

"C:\Windows\system32\notepad.exe" C:\Users\Administrator\Desktop\R.txt

File path

\Device\HarddiskVolume2\Windows\System32\notepad.exe

Executable SHA256

7d453801b059e4dab59b1b159ccd713e1d3593faa537c6ac5bbc2ce6c1e78a4d

The adversary opened a text file named **R.txt** located on the Administrator's desktop. The brief duration suggests a quick reference to instructions, passwords, or configuration parameters immediately preceding the execution of the primary payload.

Process: FalconSensor_Windows(1).exe

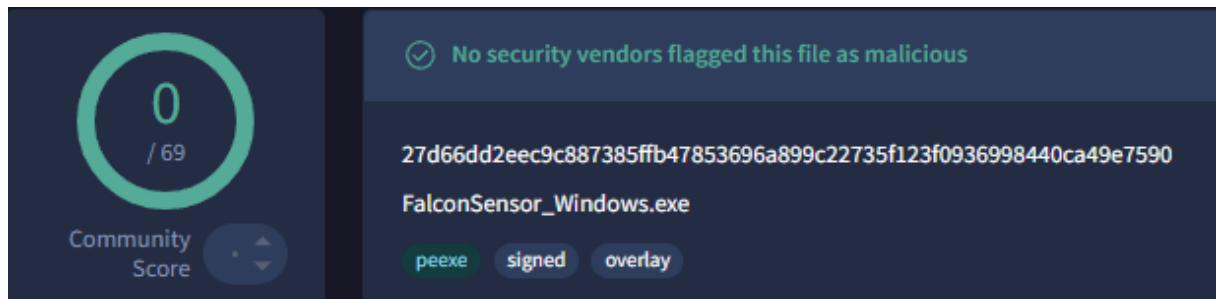
Start time: Feb. 2, 2025 06:50:49

End time: Feb. 19, 2025 06:51:00

Executable SHA256

27d66dd2eec9c887385ffb47853696a899c22735f123f0936998440ca49e7590

[Result there]<https://www.virustotal.com/gui/file/27d66dd2eec9c887385ffb47853696a899c22735f123f0936998440ca49e7590/detection>]



The adversary utilized a filename mimicking a legitimate CrowdStrike installer to avoid detection during manual inspection.



Identified by CrowdStrike as an automated lead:

Detection: Lateral Movement via Remote Desktop Protocol

Detection Timestamp: Feb. 2, 2025 06:51:55

Command line

"C:\Users\Administrator\Desktop\FalconSensor_Windows(1).exe"

File path

\Device\HarddiskVolume2\Users\Administrator\Desktop\FalconSensor_Windows(1).exe

This execution was captured as an **Automated Lead** for **Lateral Movement via Remote Desktop Protocol**. The system recognized the PE file being written and executed within the RDP session, linking it to potential manual adversary activity.

Sub-process Execution: Internal Installer Logic FalconSensor_Windows(1).exe

Start time: Feb. 2, 2025 06:50:51

End time: Feb. 2, 2025 06:51:00

Command line

```
"C:\Program Files  
(x86)\CSInstallTemp{F99C8843-97FA-491F-A473-E5EDFD42B16F}\.cr\FalconSensor_Windows(1).exe"  
-burn.clean.room="C:\Users\Administrator\Desktop\FalconSensor_Windows(1).exe" -burn.filehandle.attached=580 -burn.filehandle.self=588
```

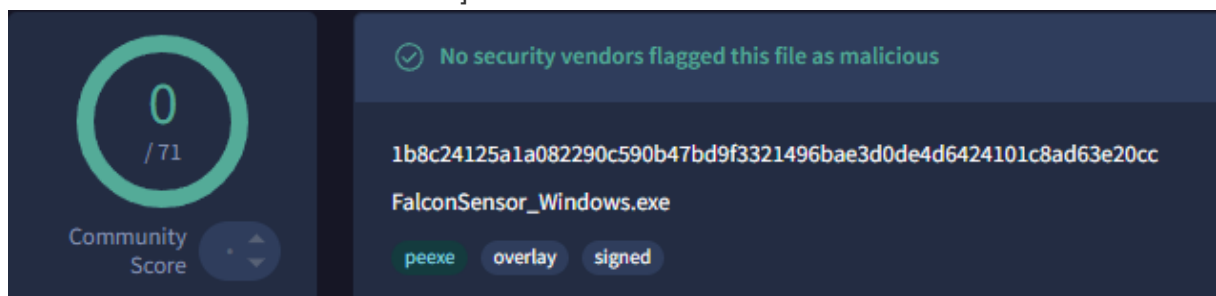
File path

```
\Device\HarddiskVolume2\Program Files  
(x86)\CSInstallTemp{F99C8843-97FA-491F-A473-E5EDFD42B16F}\.cr\FalconSensor_Windows(1).exe
```

Executable SHA256

```
1b8c24125a1a082290c590b47bd9f3321496bae3d0de4d6424101c8ad63e20cc
```

[Result there]<https://www.virustotal.com/gui/file/1b8c24125a1a082290c590b47bd9f3321496bae3d0de4d6424101c8ad63e20cc/detection>]



This child process was spawned from the desktop executable into a temporary installation directory. It utilized flags such as `-burn.filehandle.attached` and `-burn.filehandle.self`, which are common in legitimate installers but were leveraged here to further the execution of the unsigned payload.

② Phase 2 : Script-Based Execution and C2 Beaconing

Process: explorer.exe

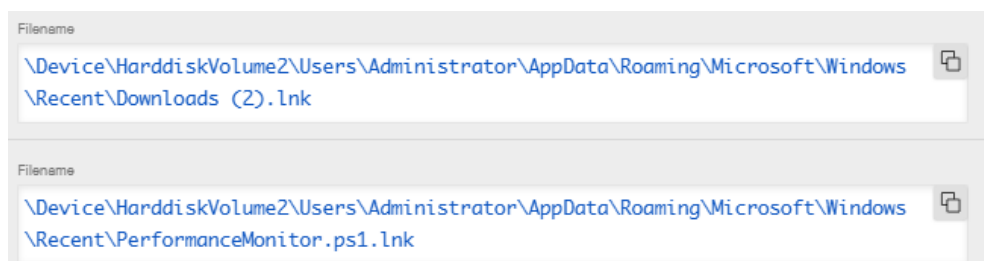
Start time: Feb. 2, 2025 09:39:47

End time: Feb. 2, 2025 09:42:45

Executable SHA256

53f36699c35c8f2360608a79f0809ba888c61f15886ae2b1f209a3e9b896cba7

The adversary used the legitimate **explorer.exe** shell to navigate the file system and initiate the second stage of the attack. Logs show the process accessing shortcut files (**.lnk**) for **PerformanceMonitor.ps1** in the Downloads folder. This confirms that the script execution was a **manual, deliberate action** performed via the interactive RDP session.



Process: powershell.exe

Start time: Feb. 2, 2025 09:40:59

End time: Feb. 2, 2025 09:42:16

Command line

```
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"  
"-Command" "if((Get-ExecutionPolicy ) -ne 'AllSigned') {  
Set-ExecutionPolicy -Scope Process Bypass }; &  
'C:\Users\Administrator\Downloads\PerformanceMonitor.ps1'"
```

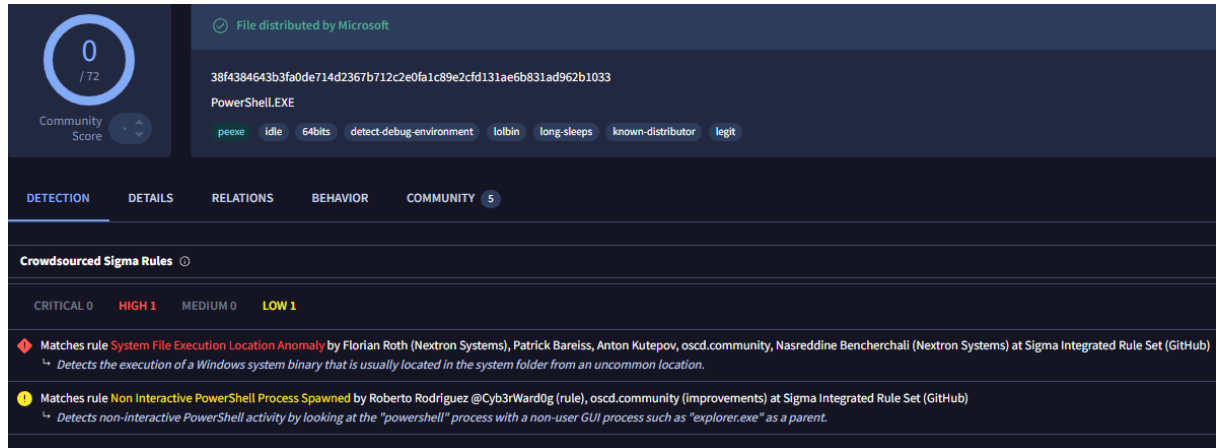
Executable SHA256

38f4384643b3fa0de714d2367b712c2e0fa1c89e2cfd131ae6b831ad962b1033

The PowerShell executable itself is legitimate. The high-severity detection is based on the **malicious command line arguments** and the script it executed (**PerformanceMonitor.ps1**), which initiated C2 communication with a known-malicious domain.

The attacker executed the **PerformanceMonitor.ps1** script with a **Bypass** flag to circumvent local security policies. This script acted as a loader for the final payload. During execution, a secondary script was written to the **\$Recycle.Bin** directory—a common **Defense Evasion** tactic used to hide artifacts from standard monitoring.

[Result there] <https://www.virustotal.com/gui/file/38f4384643b3fa0de714d2367b712c2e0fa1c89e2cfd131ae6b831ad962b1033/detection>



File distributed by Microsoft

38f4384643b3fa0de714d2367b712c2e0fa1c89e2cfd131ae6b831ad962b1033

PowerShell.EXE

peexe idle 64bits detect-debug-environment lolbin long-sleeps known-distributor legit

Community Score: 0 / 72

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 5

Crowdsourced Sigma Rules

CRITICAL 0 HIGH 1 MEDIUM 0 LOW 1

Matches rule **System File Execution Location Anomaly** by Florian Roth (Nextron Systems), Patrick Bareiss, Anton Kutevov, oscd.community, Nasreddine Bencherchali (Nextron Systems) at Sigma Integrated Rule Set (GitHub)
↳ Detects the execution of a Windows system binary that is usually located in the system folder from an uncommon location.

Matches rule **Non Interactive PowerShell Process Spawned** by Roberto Rodriguez @Cyb3rWard0g (rule), oscd.community (improvements) at Sigma Integrated Rule Set (GitHub)
↳ Detects non-interactive PowerShell activity by looking at the "powershell" process with a non-user GUI process such as "explorer.exe" as a parent.

Identified by CrowdStrike as an High-Severity Detection

File path

`\Device\HarddiskVolume2\Windows\System32\WindowsPowerShell\v1.0\powershell.exe`

Detection: Falcon Intel via Intelligence Indicator - Domain

Triggering Domain: `shichizaoa.top`

Threat Type: Cobalt Strike / RAT / Targeted

Timestamp: Feb. 2, 2025 09:42:12 - Feb. 2, 2025 09:42:14 - Feb. 2, 2025 09:43:28

Risk score: 69 **Severity:** High

Upon execution, the script initiated repeated DNS requests to the domain `shichizaoa.top`. CrowdStrike Intelligence identifies this infrastructure as a Cobalt Strike C2 server used in targeted operations. The logs recorded a consistent pattern of "Suspicious DNS Requests" (beaconing frequency), indicating the establishment of a persistent, remote command-and-control channel between the host and the adversary.

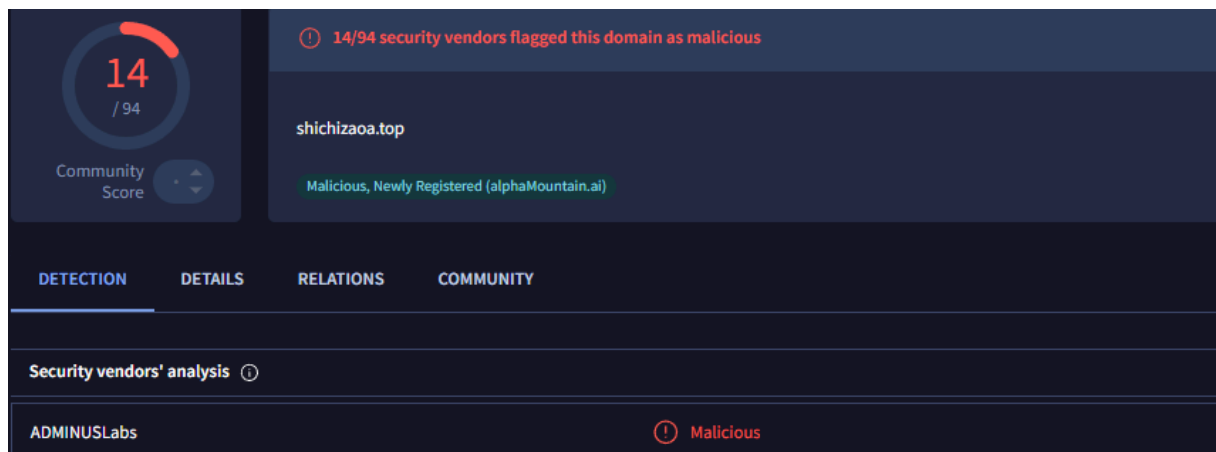
```
DnsRequests[9].DomainName: shichizaoa.top
DnsRequests[9].InterfaceIndex: 0
DnsRequests[9].LoadTime: 1770043308
DnsRequests[9].RequestType: AAAA
```

```
RiskScore: 69
Severity: 70
SeverityName: High
```

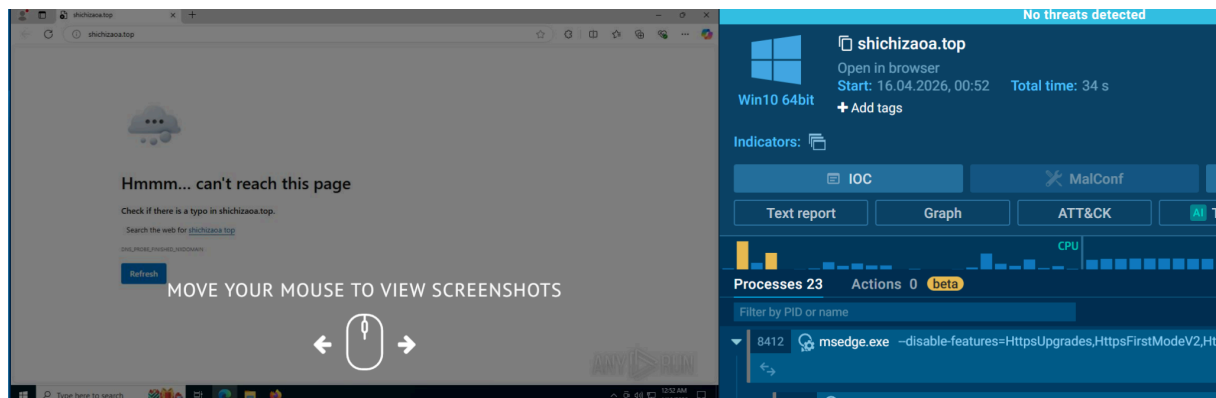
The process initiated a series of **AAAA DNS requests** for **shichizaoa.top** with a high frequency (3-5 second intervals).

The high frequency of DNS requests and the association with a **Cobalt Strike** indicator (per Falcon Intel) strongly suggests an active command-and-control session is currently in progress.

[Resultthere|<https://www.virustotal.com/gui/domain/shichizaoa.top>]



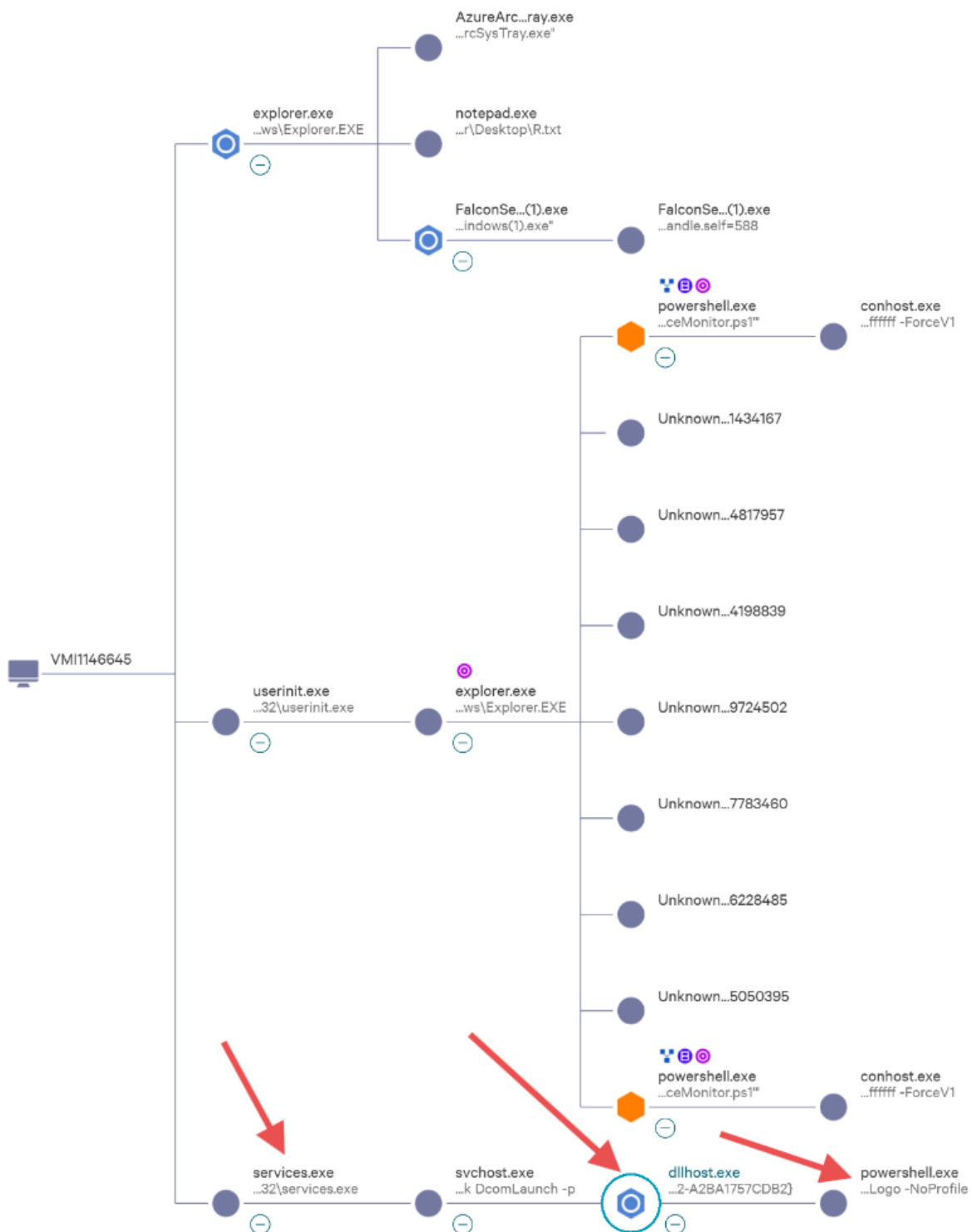
A live sandbox analysis via Any.run shows that the domain is currently unreachable. Current status of the shichizaoa.top domain. The infrastructure is no longer active, which is consistent with "burner" C2 domains used in targeted attacks to evade post-incident forensic analysis.



Rawstring :

```
@rawstring:{"ProcessStartTime":1738043259,"ProcessEndTime":0,"ProcessId":6047740540713,"ParentProcessId":6047673485759,"Hostname":"[redacted-domain]","UserName":"Administrator","Name":"Intel_Detection","Description":"A domain lookup matched a CrowdStrike Intelligence indicator that has previously been used in targeted attacks.","Severity":70,"SeverityName":"High","FileName":"powershell.exe","FilePath":"\\\\Device\\HarddiskVolume2\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe","CommandLine":"\"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\" \"-Command\" \"if((Get-ExecutionPolicy) -ne 'AllSigned') { Set-ExecutionPolicy -Scope Process Bypass }; \u0026 'C:\\Users\\Administrator\\Downloads\\PerformanceMonitor.ps1'\"", "SHA256String": "38f4384643b3fa0de714d2367b712c2e0fal89e2cfd131ae6b831ad962b1033", "MD5String": "dd6f4b7818a253887b8ea86515f6fb7d", "SHA1String": "00000000000000000000000000000000", "LogonDomain": "[redacted-domain]", "FalconHostLink": "(internal-link, redacted-for-confidentiality)", "DnsRequests": [{"DomainName": "shichizaoa.top", "RequestType": "AAAA", "LoadTime": 1738043268, "InterfaceIndex": 0}, {"DomainName": "shichizaoa.top", "RequestType": "AAAA", "LoadTime": 1738043268, "InterfaceIndex": 0}, {"DomainName": "shichizaoa.top", "RequestType": "AAAA", "LoadTime": 1738043273, "InterfaceIndex": 0}, {"DomainName": "shichizaoa.top", "RequestType": "AAAA", "LoadTime": 1738043278, "InterfaceIndex": 0}, {"DomainName": "shichizaoa.top", "RequestType": "AAAA", "LoadTime": 1738043283, "InterfaceIndex": 0}, {"DomainName": "shichizaoa.top", "RequestType": "AAAA", "LoadTime": 1738043288, "InterfaceIndex": 0}, {"DomainName": "shichizaoa.top", "RequestType": "AAAA", "LoadTime": 1738043294, "InterfaceIndex": 0}, {"DomainName": "shichizaoa.top", "RequestType": "AAAA", "LoadTime": 1738043298, "InterfaceIndex": 0}, {"DomainName": "shichizaoa.top", "RequestType": "AAAA", "LoadTime": 1738043302, "InterfaceIndex": 0}, {"DomainName": "shichizaoa.top", "RequestType": "AAAA", "LoadTime": 1738043308, "InterfaceIndex": 0}], "IOType": "domain", "IOCValue": "shichizaoa.top", "CompositeId": "55ff35c57f0441f19baad0a47c239f7d:ind:fd0ae32b624a4baa83845321c1cf0e52:6047740540713-4900-1525008", "LocalIP": "[redacted-internal-IP]", "MACAddress": "00-50-56-49-71-48", "Tactic": "Falcon Intel", "Technique": "Intelligence Indicator - Domain", "Objective": "Falcon Detection Method", "PatternDispositionDescription": "Detection, standard detection.", "PatternDispositionValue": 0, "PatternDispositionFlags": {"Indicator": false, "Detect": false, "InddetMask": false, "SensorOnly": false, "Rooting": false, "KillProcess": false, "KillSubProcess": false, "QuarantineMachine": false, "QuarantineFile": false, "PolicyDisabled": false, "KillParent": false, "OperationBlocked": false, "ProcessBlocked": false, "RegistryOperationBlocked": false, "CriticalProcessDisabled": false, "BootupSafeguardEnabled": false, "FsOperationBlocked": false, "HandleOperationDowngraded": false, "KillActionFailed": false, "BlockingUnsupportedOrDisabled": false, "SuspendProcess": false, "SuspendParent": false, "ContainmentFilesystem": false}, "ParentImageFileName": "explorer.exe", "ParentCommandLine": "C:\\Windows\\Explorer.EXE", "GrandParentImageFileName": "userinit.exe", "GrandParentCommandLine": "C:\\Windows\\system32\\userinit.exe", "HostGroups": "c6cffeld95944fe48f105a75c4bf4ec6", "PatternId": 4900, "SourceVendors": "CrowdStrike", "SourceProducts": "Falcon Insight", "DataDomains": "Endpoint", "AggregateId": "aggind:fd0ae32b624a4baa83845321c1cf0e52:438088692920", "Type": "ldt", "ParentImagePath": "\\\\Device\\HarddiskVolume2\\Windows\\explorer.exe", "GrandParentImagePath": "\\\\Device\\HarddiskVolume2\\Windows\\System32\\userinit.exe", "LocalIPv6": "", "PlatformId": 0, "PlatformName": "Windows", "MitreAttack": [{"Tactic": "Falcon Intel", "TacticID": "CSTA0007", "Technique": "Intelligence Indicator - Domain", "TechniqueID": "CST0018", "PatternID": 4900}], "CloudIndicator": false, "RiskScore": 69, "CustomerIdString": "55ff35c57f0441f19baad0a47c239f7d", "UTCTimestamp": 1738043332, "Nonce": 14443603536944362234, "AgentIdString": "fd0ae32b624a4baa83845321c1cf0e52", "cid": "55ff35c57f0441f19baad0a47c239f7d", "eid": 118, "timestamp": "2025-02-02T14:42:12Z", "EventType": "Event_ExternalApiEvent", "ExternalApiType": "Event_EppDetectionSummaryEvent"}
```

Technical validation via Advanced Event Search confirmed a persistent C2 channel. Raw telemetry revealed a **fixed 5-second beaconing interval** and the use of **IPv6 (AAAA) DNS queries** to evade standard IPv4-centric monitoring. Furthermore, comparing PIDs across multiple logs confirmed a **re-execution mechanism**, proving the threat's persistence within the user session.

Observed Command 3:

③ Phase 3: Persistence and Privilege Escalation

Process: services.exe and svchost.exe

Start time: Feb. 2, 2025 09:43:27

Executable SHA256 (**services.exe**)

61480752127c47ea71a61217dd2d46fda104b0b48af9905ded6c607b946da3d1

[Result there]<https://www.virustotal.com/gui/file/61480752127c47ea71a61217dd2d46fda104b0b48af9905ded6c607b946da3d1>]

Executable SHA256 (**svchost.exe**)

d7f10571f58adbe122f615b869eb42b05286770148067cbf88054c38f1bca4c4

[Result there]<https://www.virustotal.com/gui/file/d7f10571f58adbe122f615b869eb42b05286770148067cbf88054c38f1bca4c4>]

Both hashes were verified against public repositories and confirmed as legitimate Microsoft binaries.

The attack has progressed to the stage involving system services. The fact that the execution chain originates from “**services.exe**” confirms that the attacker created or modified a system service to automatically launch malicious code with **SYSTEM** privileges.

[redacted-host]

By registering a malicious service or hijacking a COM component, the attacker ensured the payload would execute automatically under the LocalSystem account, bypassing user-level restrictions.

Process: dllhost.exe

Start time: Feb. 3, 2025 18:07:26

End time: Feb. 3, 2025 18:08:01

Executable SHA256

0fa66b69bf46300473fd37147fd0fe9d618d0f8d33a6987cdd65bcbdd3a68e87e

[Result there]<https://www.virustotal.com/gui/file/0fa66b69bf46300473fd37147fd0fe9d618d0f8d33a6987cdd65bcbdd3a68e87e/relations>]

While the binary is authentic, it was leveraged by the adversary to host a malicious COM object. This allowed the attacker to execute code within a trusted system process, a technique known as System Binary Proxy Execution. The Debugger Evasion alert confirms that the code loaded into **dllhost.exe** performed environmental checks to detect analysis tools before launching the final PowerShell payload.

Identified by CrowdStrike as an automated lead:

CrowdStrike detected **Debugger Evasion**, indicating that the malware actively checked for signs of analysis attempts before launching its main module. This behavior is characteristic of advanced RATs and Cobalt Strike beacons.

Detection: Defense evasion via Debugger Evasion

Detection Timestamp: Feb. 3, 2025 18:09:00

Command line

```
C:\Windows\system32\DllHost.exe  
/Processid:{BD07DDB9-1C61-4DCE-9202-A2BA1757CDB2}
```

File path

```
\Device\HarddiskVolume2\Windows\System32\dllhost.exe
```

CrowdStrike detected Debugger Evasion, indicating that the malware actively checked for signs of analysis attempts before launching its main module. This behavior is characteristic of advanced RATs and Cobalt Strike beacons.

Process: powershell.exe

Start time: Feb. 3, 2025 18:07:27

End time: Feb. 3, 2025 18:07:56

Command line

```
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"  
-Version 5.1 -s -NoLogo -NoProfile
```

File path

```
\Device\HarddiskVolume2\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
```

Executable SHA256

```
38f4384643b3fa0de714d2367b712c2e0fa1c89e2cfd131ae6b831ad962b1033
```

[Result there|<https://www.virustotal.com/gui/file/38f4384643b3fa0de714d2367b712c2e0fa1c89e2cfd131ae6b831ad962b1033/detection>]

This is a highly suspicious command. The “-s” flag and the absence of a profile (“-NoProfile”) indicate that PowerShell is being utilized as a “stealthy” engine to execute commands received from the network or the registry. This process serves as a direct continuation of the C2 connection established during the second phase.

Analysis

Command Line Breakdown: Phase 1 (Initial Access & Defense Impairment)

- ♦ `explorer.exe` → Leveraged as the primary interactive shell for the RDP session. The process was flagged for requesting an cross-process handle to the **Falcon sensor** with excessive permissions. This indicates a targeted attempt to **Impair Defenses** by attempting to access the security agent's memory.
- ♦ `"C:\Windows\system32\notepad.exe" C:\Users\Administrator\Desktop\R.txt` → Used to access a text file on the Administrator's desktop. This behavior is typical of manual intrusions where the adversary references a "playbook," stored credentials, or staging instructions.
- ♦ `"C:\Users\Administrator\Desktop\FalconSensor_Windows(1).exe"` → A malicious executable manually delivered via RDP. It utilizes **Masquerading**, mimicking the naming convention of a legitimate CrowdStrike installer to deceive analysts during manual process reviews.
- ♦ `"C:\Program Files (x86)\CSInstallTemp\...\FalconSensor_Windows(1).exe"` → A child process spawned into a temporary directory using `-burn` flags. This represents the internal "dropper" logic used to extract and execute the secondary payload while attempting to appear as a standard software installation.

Command Line Breakdown: Phase 2 (C2 Loader)

Command: `powershell.exe -Command "if((Get-ExecutionPolicy) -ne 'AllSigned') { Set-ExecutionPolicy -Scope Process Bypass }; & 'C:\Users\Administrator\Downloads\PerformanceMonitor.ps1'"`

- ♦ `powershell.exe` → Execution of a legitimate interpreter (SHA256: 38f4384643b3fa0de714d2367b712c2e0fa1c89e2cfd131ae6b831ad962b1033). The use of a trusted system process to conceal malicious activity (Living off the Land).
- ♦ `-Command` → An instruction to execute the subsequent line of code without opening an interactive window.
- ♦ `if((Get-ExecutionPolicy) -ne 'AllSigned') { Set-ExecutionPolicy -Scope Process Bypass } →` Security bypass logic. The script checks the current execution policy and forcibly sets the mode to "Bypass" for the current process. This allows for the execution of the unsigned malicious file `'PerformanceMonitor.ps1'`, thereby bypassing system security settings.
- ♦ `& 'PerformanceMonitor.ps1'` → The call operator (`&`) executes the script from the Downloads folder. The "Downloads" location, combined with execution under the "Administrator" account via RDP, confirms that the payload was manually delivered by an attacker.

Command Line Breakdown: Phase 3 (Persistent C2 Backdoor)

Command: `powershell.exe -Version 5.1 -s -NoLogo -NoProfile`

- ♦ **-Version 5.1** → Forces the execution of a specific engine version. This is frequently used to ensure the compatibility of malicious code or to bypass advanced logging features (such as Script Block Logging) present in newer versions.
- ♦ **-s** → A critical indicator. This flag signifies that PowerShell is expecting commands from the standard input stream rather than from a file. In this specific chain (`dllhost.exe -> powershell.exe`), it indicates that commands are being passed directly from memory or via named pipes. This constitutes a highly stealthy method that leaves no traces on disk (a "fileless" attack).
- ♦ **-NoLogo & -NoProfile** → Suppresses startup banners and ignores user profiles. This accelerates startup, prevents conflicts, and helps the process "blend in" with background system tasks.

Potential Impact

Execution Policy Bypass: The use of `Set-ExecutionPolicy -Scope Process Bypass` successfully disabled script-signature enforcement for the PowerShell sessions. This allowed the unsigned `PerformanceMonitor.ps1` to execute and establish communication with external infrastructure, bypassing default Windows security configurations.

Malicious C2 Communication (Cobalt Strike): The high-frequency beaconing to `shichizaoa.top` (detected as Cobalt Strike) confirms the host was successfully compromised by a professional post-exploitation framework. This provides the adversary with a persistent "remote shell," enabling real-time command execution, data exfiltration, and internal reconnaissance.

Full System Compromise (Privilege Escalation): The transition from an `Administrator` session to the LocalSystem (SYSTEM) context via `services.exe` granted the adversary unrestricted control over the host. At this privilege level, the attacker can access sensitive system files, modify the registry, and potentially disable security sensors entirely.

Active Persistence: Unlike transient attacks, this intrusion successfully established Persistence. By hijacking a COM component (`dllhost.exe`) and registering it as a system service, the adversary ensured the malicious beacon would automatically restart upon reboot, surviving user logoffs and session terminations.

Living-off-the-Land (LotL) Success: The adversary successfully utilized legitimate binaries (`powershell.exe`, `dllhost.exe`, `explorer.exe`) to perform unauthorized actions. This strategy was intended to blend malicious activity with normal system operations, significantly complicating detection and increasing the "dwell time" of the threat actor within the network.

Defense Evasion & Debugger Checks: The detection of Debugger Evasion indicates the payload was equipped with anti-analysis features. This suggests the adversary is sophisticated and actively taking steps to prevent forensic investigators from tracing or reverse-engineering their toolset.

Response

A review of the raw telemetry via Advanced Event Search (AES) confirmed that no automated mitigation was performed by the sensor. Specifically, the **PatternDispositionFlags** for **KillProcess**, **QuarantineFile**, and **OperationBlocked** were all identified as **false**.

```
PatternDispositionFlags.KillActionFailed: false
PatternDispositionFlags.KillParent: false
PatternDispositionFlags.KillProcess: false
PatternDispositionFlags.KillSubProcess: false
PatternDispositionFlags.OperationBlocked: false
PatternDispositionFlags.PolicyDisabled: false
PatternDispositionFlags.ProcessBlocked: false
PatternDispositionFlags.QuarantineFile: false
PatternDispositionFlags.QuarantineMachine: false
```

Although the malicious script **PerformanceMonitor.ps1** (and the identified dropper **FalconSensor_Windows(1).exe**) was not found on the disk during manual inspection, this is not a result of EDR intervention.

The absence of the files suggests a **self-deletion routine** or **manual cleanup by the adversary** as part of their **Defense Evasion** strategy. Because the operation was not blocked, the adversary was able to successfully complete the execution chain, including the establishment of a C2 beacon and the creation of persistent system services.

Overall Analyst Assessment

An interactive RDP session was established on host [redacted-host], where an adversary manually initiated a multi-stage intrusion. The attack began with efforts to **Impair Defenses** against the Falcon sensor and the execution of a masqueraded dropper, **FalconSensor_Windows(1).exe**. Subsequently, the adversary executed **PerformanceMonitor.ps1** from the Downloads folder using a PowerShell execution-policy bypass to establish a command-and-control (C2) channel.

CrowdStrike identified the outbound traffic as **Cobalt Strike / RAT** activity targeting the domain **shichizaoa.top**. Advanced telemetry confirmed a persistent, automated **beaconing pattern** with a fixed 5-second interval and the use of IPv6 (AAAA) queries to evade network detection. Unlike transient script execution, this incident resulted in successful **Privilege Escalation** to the **SYSTEM** context and the establishment of **Persistence** via the Service Control Manager and **dllhost.exe**.

While the EDR provided high-severity detections, **no automated mitigation or quarantine occurred** (Detection Only). The subsequent absence of malicious files on the disk, combined with detected **Debugger Evasion**, confirms a sophisticated adversary utilizing anti-forensic techniques to scrub evidence of a successful compromise.

This incident demonstrates a complete and successful attack chain—from initial manual RDP access to high-privilege persistence. Given that the C2 communication was not blocked and the malicious service remains active, the host must be considered **fully compromised** and requires immediate isolation and forensic remediation.

Mini Attack Visualization

[Administrator logs in via RDP (Initial Access)]



[explorer.exe requests handle to Falcon sensor process (Impair Defenses)]



[notepad.exe opens R.txt on Desktop (Adversary playbook/instruction reference)]



[FalconSensor_Windows(1).exe executed from Desktop (T1036.005 – Masquerading)]



[explorer.exe interacts with .lnk files in Downloads → spawns PowerShell with bypass]



[PerformanceMonitor.ps1 executed → initiates Cobalt Strike C2 beaconing to shichizaoa.top]



[Outbound AAAA DNS requests observed (IPv6 evasion) with fixed 5-second jitter]



[Script writes artifacts to \$Recycle.Bin → adversary performs manual self-deletion/cleanup]



[services.exe triggers persistence → launches dllhost.exe under SYSTEM context (PrivEsc)]



[dllhost.exe performs Debugger Evasion → spawns stealth PowerShell (-s stdin mode)]



[CrowdStrike Falcon: Detection Only (No Prevention) — C2 established; Persistence active]



[Host [redacted-host] fully compromised — Persistence survives user logoff]

Recommended actions

1. Host Isolation & Verification

- **Isolate the Host:** Immediately perform **Network Containment** on [redacted-host] via the CrowdStrike console to sever the active C2 beacon.
- **Search and Remove:** Search for and delete **PerformanceMonitor.ps1** from **C:\Users\Administrator\Downloads** and **FalconSensor_Windows(1).exe** from the **Desktop**.
- **Recycle Bin Audit:** Thoroughly inspect and purge the **\$Recycle.Bin** directory for hidden or secondary script artifacts.

2. Malware & Persistence Cleanup

- **Service & COM Removal:** Identify and delete the malicious system service or COM object registration created via **services.exe** that triggers **dllhost.exe**.
- **Process Termination:** Manually terminate active malicious process IDs: **8636, 2512, 4752** (**powershell.exe**) and **8248** (**dllhost.exe**).
- **Fileless Audit:** Since the attack utilized the **-s** (stdin) flag, perform a memory scan to ensure no malicious code remains resident in other system processes.

3. Network & IOC Controls

- **Domain Blocking:** Blacklist and monitor all outbound connections to the C2 domain **shichizaooa.top** at the perimeter firewall.
- **IOC Hunt:** Add the dropper hash **27d66dd2eec9c887385ffb47853696a899c22735f123f0936998440ca49e7590** to the global blacklist.
- **Detection Tuning:** Create custom IOA rules to alert on **dllhost.exe** spawning **powershell.exe**, especially with the **-s** (stdin) or **-Version 5.1** flags.

4. Credential & Account Security

- **Password Reset:** Force an immediate password reset for the **Administrator** account.
- **RDP Audit:** Review RDP logs for the last 48 hours to identify the source IP of the initial connection and check for any other compromised hosts.
- **Privilege Review:** Audit local administrator groups and system-level permissions to ensure no other backdoors were created during the **SYSTEM** context phase.

5. PowerShell & System Hardening

- **Execution Policy:** Enforce a strict **AllSigned** execution policy across the environment.
- **Logging:** Ensure **PowerShell ScriptBlock Logging** is enabled to capture future fileless/stdin commands.
- **COM Security:** Restrict the ability of non-standard services to interact with COM+ surrogates (**dllhost.exe**) unless explicitly required.

6. Escalation

- **Incident Response:** Escalate this ticket to the **Level 3 / Incident Response (IR) team** immediately. Due to the successful establishment of a **SYSTEM-level beacon**, this incident must be treated as a full compromise of the server.